MILITARY SPECIFICATION

AMPLIFIER-INDICATOR GROUP, ALTITUDE-VERTICAL
SPEED A/A24G-7

This amendment forms a part of MIL-A-27328(USAF), dated 14 Jan 60, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Delete "2.1" and add: "2.1 Government documents."

* Add:
"2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

* Under SPECIFICATIONS, Federal, delete "QQ-P-416" and add:

"PPP-B-601 Box, Wood, Cleated Plywood"
"PPP-B-636 Boxes, Shipping, Fiberboard".

Under SPECIFICATIONS, Military: delete "MIL-E-5272", "MIL-E-7894", "MIL-P-7936", "MIL-T-9107", "MIL-I-26600" and "MIL-D-70327". Add:

"DOD-D-1000 Drawings, Engineering and Associated Lists
MIL-C-83488 Coating, Aluminum, ION Vapor Deposited"

Add boxed paragraph to bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

* Under Standard, Military, Delete "MS33586" and add:

"MIL-STD-143 Standards and Specifications, Order Of Precedence For The Selection Of
MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization Of
MIL-STD-794 Parts and Equipment Procedures For Packaging and Packing Of
* MIL-STD-810 Environmental Test Methods

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MIL-STD-826 Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-889 Dissimilar Metals"

Delete "Publications" and document referenced there under.

* "2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* "3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.5 and 6.2c)."

3.4 Line 3: Delete "ANA Bulletin 143" and add: "MIL-STD-143".

PAGE 3

* Add:

"3.5.1.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item."

3.5.2.1 Line 4: Delete "MS33586 and add: "MIL-STD-889"

PAGE 4

3.6.3 Line 2: Delete "MIL-E-7894" and add: "MIL-STD-704". Line 8 and 9: Delete "cps" and add: "Hz" in both places.

PAGE 10

Figure 6: Delete and add the new figure 6 in this amendment.

PAGE 12

* 3.7 Step a; Delete and add: "In accordance with Method 514.2 of MIL-STD-810.".

* Step b; Delete and add: "In accordance with Method 513.2 of MIL-STD-810.".

* Step h; Delete and add: "In accordance with Method 516.2 of MIL-STD-810.".

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PAGE 14

3.7.2 Line 4: Delete "MIL-I-26600" and add: "MIL-STD-826".

3.8 Line 5: Delete "MIL-D-70327" and add: "DOD-D-1000".

PAGE 15

3.9.3. Line 1: Delete "cps" and add: "Hz".

PAGE 19

3.10.1 Line 3: Delete "1" and add: "1.62".

3.10.2 Line 2: Delete "1" and add: "2".

3.10.4 Add "(Hz)" following "Bandwidth".

3.10.4.1 Line 7: Delete "cps" and add "Hz".

3.10.5 Line 2: Between "permit" and "adjustment" insert "minimum".

PAGE 20

3.14.2 Delete existing paragraph and add:

"3.14.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

PAGE 24

4.1, b: Delete "Preproduction" and add: "First article".

PAGE 25

4.2.1.1 f: Immediately following "Lighting" add "(visual comparison)".

PAGE 26

* 4.2.2.1.1 Step d; Delete "procedure IV".

4.2.2.1.1 Add: "g. Stray light".

* 4.2.2.2.1 Step h; Delete "Salt spray" and add: "Salt fog".

* 4.2.2.2.2 Step g; Delete "Salt spray" and add: "Salt fog".

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4.3.1.4 Line 2: Delete "cps" and add: "Hz".

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4.3.1.4. Line 3: Add "dc" following "28V ±2 percent".

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4.3.1.9 Delete "cps" and add: "Hz".

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4.3.2.3.1 Line 2: Delete "cps" and add: "Hz".

4.3.2.4 Line 1: Delete "cps" and add: "Hz".

PAGE 33

4.3.2.5 Line 2: Delete "cps" and add: "Hz".

4.4.1.2 Line 1: Delete first sentence and add: "The functional check shall be performed prior to final sealing and final inspection of the indicator."

PAGE 34

4.4.1.5 Line 2: Delete "cps" and add: "Hz".

PAGE 41

* 4.4.1.14 Lines 2 and 3; Delete "procedure IV of MIL-E-5272 at amplitudes of 0.009 to 0.011 inch" and add: "Method 514.2, procedures I, part 1 (curve c), category b.1 equipment of MIL-STD-810C. For this test, only the sinusodial cycling portions will be required. The cycling and curve c will be modified as follows: 'Cycling time shall be the time of one sweep time. A range for curve c of 5 to 50 Hz and an acceleration level of ±1g from 23 to 50 Hz.'".

PAGE 42

Figure 11 Delete and add: the new figure 11 in this amendment.

PAGE 44

4.4.1.18 Delete existing paragraph and add:
"The indicator shall be subjected to a vibration test in accordance with Method 514.2, procedure I, part 1, (curve c), category b.1 equipment of MIL-STD-810. The frequency of vibration shall be varied uniformly at a sweep time of one hour. Curve c shall be modified as follows: 'A range for curve c of 5 to 50 Hz and an acceleration level of ±1g from 23 to 50 Hz.'"

4.4.1.19 Delete the last sentence and substitute the following: "After being returned to sea level pressure, the indicator shall meet the requirements of the individual tests, paragraph 4.2.1.1".

* 4.4.1.20 Delete existing paragraph and add: "The indicator shall be subjected to an acceleration test in accordance with Method 513.2, procedure II of MIL-STD-810C. The forward acceleration level (A) shall be 1.5g and produced by means of a centrifuge facility. The indicator is used on fighter aircraft, therefore G test levels from table 513.2-II must be modified accordingly.".

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* 4.4.1.21 Line 4: Delete "procedure I of MIL-E-5272" and add: "Method 507.1, procedure I of MIL-STD-810C.".

* 4.4.1.23 Delete existing title "Salt Spray" and add: "Salt Fog". Delete on Line 4 "Salt Spray Test in accordance with MIL-E-5272" and add: "Salt Fog test in accordance with Method 509.1, procedure I of MIL-STD-810C.".

4.4.1.24 Line 6: Delete "2 hours" and add: "1 hour".

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4.4.1.24 Lines 4 and 5: Under "Room Temp" delete "-55°C and +70°C" and add: "-10°C and +50C" respectively.

Endurance Test Table: Under the column "Ambient Temperature °C" delete "70" and add: "50" in three places and delete "-55" and add: "-10" in two places.

Delete "4.4.1.24 and add:

* 4.4.1.24 Endurance. The indicator and amplifier shall be tested for endurance for at least 700 hours of electrical operation with a cycling time for each indicator function of 6 to 7 minutes per cycle (total of at least 6000 cycles per function). The test will begin when rated power is applied. After 4 hours of rated power on, the power shall remain off for a period of 2 hours. This procedure of power-on power-off operation shall be repeated until the 700 hours of actual electrical operations are achieved. after each 50 hours of electrical operation, the indicator shall be subjected to a scale error test of at least 3 test stations per function. Room tolerances shall apply throughout. The allowable tolerance shall be increased by 50 percent. The conditions under which the systems shall be tested are listed in the table below. The table represents the following percentage breakdown of test time.

Room Temp.	64 percent
-10°C	14 percent
+50°C	22 percent

Endurance Test Table

Cycling Period	Number of Hours	Ambient Temperature °C
1	50	25
2	50	50
3	100	25
4	50	-10
5	50	25
6	50	50
7	100	25
8	50	-10
9	100	25
10	50	50
11	50	25

During the fifth cycle the indicator shall be vibrated as specified in 4.4.1.14

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except as follows: 'The indicator shall be vibrated at a frequency of 35 Hz at a double amplitude of 0.036 inch in lieu of curve C.' The transition from the fourth to fifth cycles shall be completed in the shortest possible time allow the vibration test to be conducted during temperature change from $-10^{o}C$ to $+25^{o}C$. In case of malfunction of any kind, the malfunction shall be recorded and the life test interrupt while repairs are being made. The life test shall then be continued from point of interruption. Failure to meet scale error tests shall be considered a malfunction and so recorded. Upon completion of of the life test the indicator will again be tested for scale error. If recalibration is required to pass the scale error test an additional malfunction shall be charged against the life test. Upon completion of the test the mean life shall be determined from the following equation.

$$\text{Mean life} - \frac{700N}{M}$$

Where N = number of units under test
M = number of malfunctions
The mean life of the indicator shall be at least 500 hours.

PAGE 50

4.4.2.9 Line 2 and 4: Delete "cps" and add: "Hz" in both places.

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4.4.2.10.3.3 Line 3: Delete "J1202U" and add: "J1201U".

PAGE 58

* 4.4.2.13 Delete sentences 1 and 2 and add: "The unit shall be subjected to a vibration test in accordance with Method 514.2, procedure I, part 2, category b.1 equipment of MIL-STD-810C, except as follows: 'A .4 inch double amplitude from 5 to 7 Hz and a maximum of +1g acceleration from 7 to 500 Hz in lieu of Figure 514.2-2 and given curve.' For this test, only the sinusoidal cycling portion will be required.".

* Line 3: Delete "I" and add: "XII". Lines 4 and 5: Delete "cps" and add: "Hz" in both places.

* 4.4.2.16 Line 2: Delete "MIL-E-5272, procedure II." and add: "Method 516.2, procedure I and III of MIL-STD-810C. Shock pulse shape shall be in accordance with Figure 516.2-2 (amplitude a and time duration c) for both procedures. The amplifier shall be operated during procedure I.".

* 4.4.2.17 Delete first sentence and add: "The unit shall be subjected to an acceleration test in accordance with Method 513.2, procedures II of MIL-STD-810C. The forward acceleration level shall be 2g and produced by means of a centrifuge facility. The amplifier is used on fighter aircraft, therefore G test levels for table 513.2-II must be modified accordingly.".

* 4.4.2.18 Delete "MIL-E-5272, procedure I" and add: "Method 507.1, procedure I of MIL-STD-810C."

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4.4.2.19 Line 2: Delete "MIL-E-5400" and add: "MIL-STD-826". Delete the last sentence and add: "The conducted noise requirements shall not apply to those conductors which interconnect components."

* 4.4.1.20 Delete existing title "Salt Spray" and add: "Salt Fog". Delete on Line 1 & 2, "salt spray test in accordance with MIL-STD-5272." and add: "salt fog test in accordance with Method 509.1, procedure I of MIL-STD-810C.".

* 4.4.2.21 Line 2: Delete of "MIL-E-5272." and add: "in accordance with Method 507.1, procedure I of MIL-STD-810c.".

4.5 through 4.5.4 Delete "Preproduction" wherever used and add: "First Article".

* Delete "4.5.2".

PAGE 60

Add the following paragraph to section 4:

4.6 Inspection of packaging. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

* Delete Section 5 in its entirety and add:

"5. PACKAGING.

5.1 Preservation-Packaging. Preservation-packaging shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Amplifier shall be cleaned in accordance with C-1 of MIL-P-116.

5.1.1.2 Drying. Amplifier shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation Application. Not applicable.

5.1.1.4 Unit Packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IA-8 of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the amplifier when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Amplifier shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration,

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and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of amplifier shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Amplifiers packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Amplifiers packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged amplifier shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129."

6.2 c Delete "preproduction" and add: "first article".

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Add new paragraph 6.3 as follows:

"6.3 First article. First article tests and inspection are required. First articles shall be tested and approved under appropriate provisions of 7-104.55 of the Defense Aquisition Regulation. First article tests are described in 4.5. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Air Force - 71

Project Number:
6610 - F129

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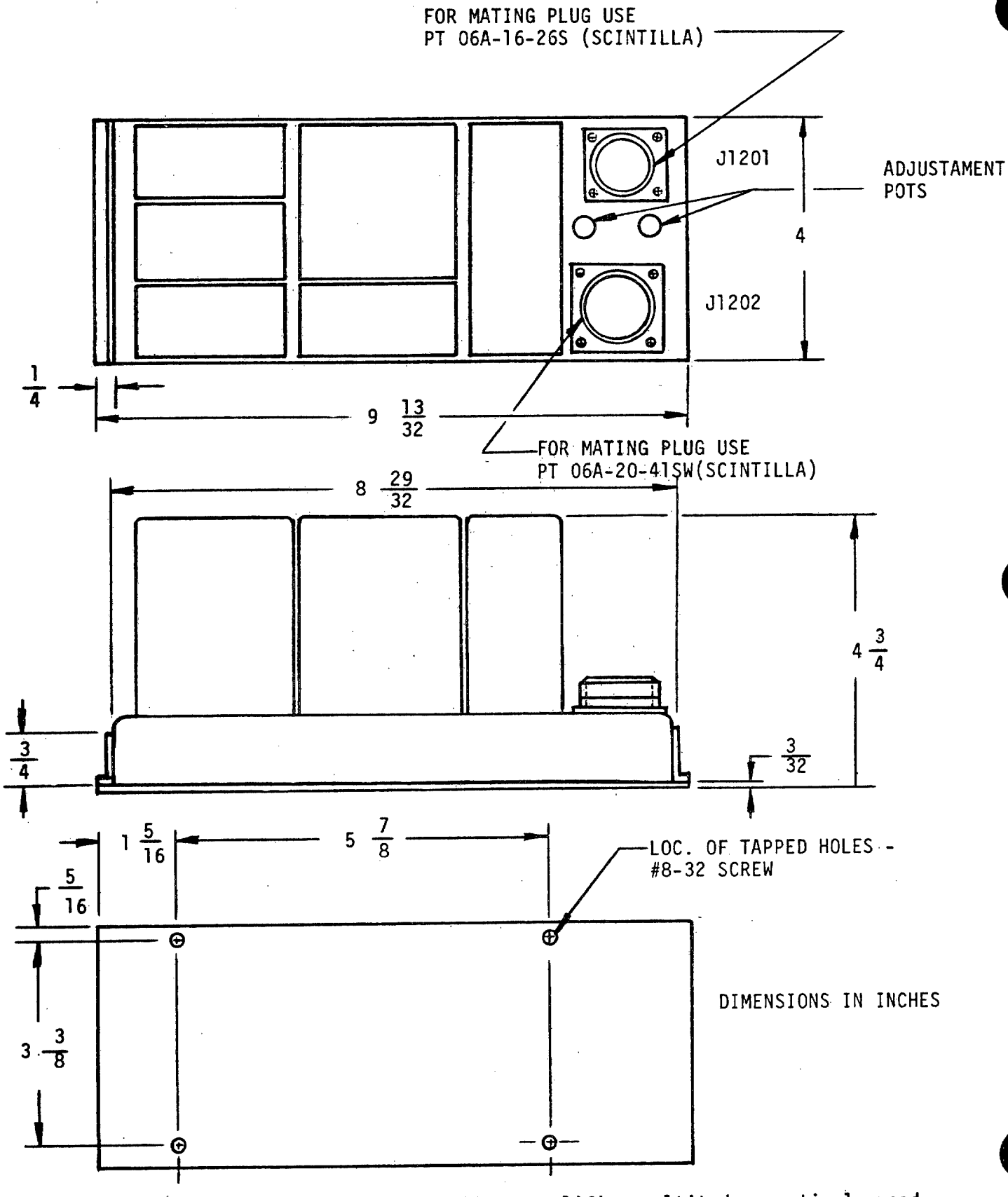


FIGURE 6. Outline, amplifier, altitude, vertical speed.

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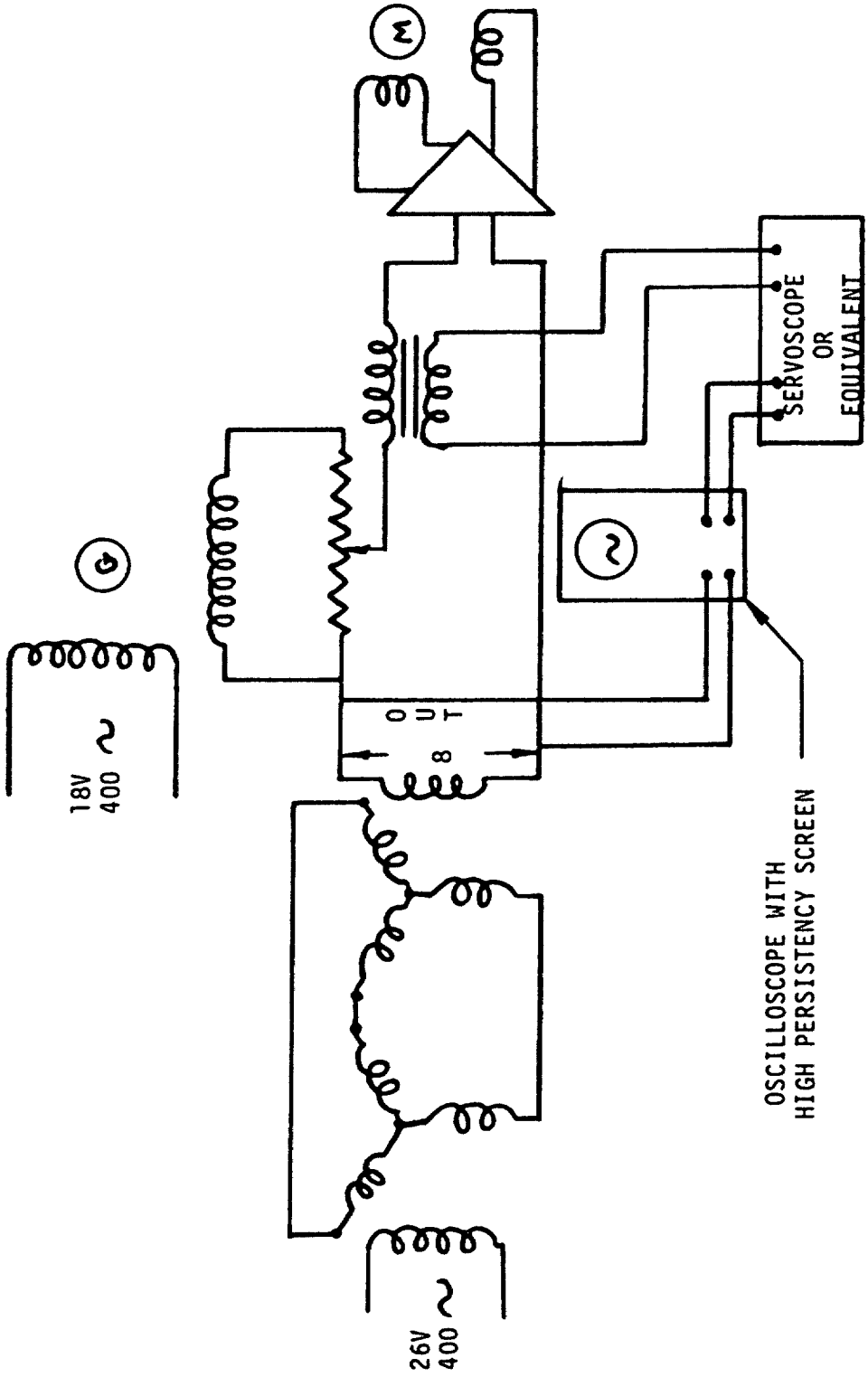


FIGURE 11. Response test circuit.

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